Yasra Chandio

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Department of Electrical and Computer Engineering
University of Massachusetts Amherst
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Education

PhD in Computer Engineering, University of Massachusetts Amherst. 2020–2025

MS in Computer Science, Binghamton University (SUNY). 2018–2019

BS in Software Engineering, Mehran University of Engineering and Technology, Pakistan *[Bronze]* 2010–2014

Awards Highlights

MIT EECS Rising Star | 19% acceptance rate | Link 2024
— *Highly selective workshop based on academic excellence and commitment to advancing DEI.*

Cyber-Physical Systems (CPS) Rising Star | 16% acceptance rate | Link 2024
— *Highly selective workshop based on outstanding achievements and future potential in CPS.*

UMASS College of Engineering Diversity, Equity, and Inclusion (DEI) Award | Link 2024
— *Award recognizing one undergraduate/graduate student/postdoc recognizing contributions to DEI.*

Research Featured in National & International News | BBC, UMASS, CoE, ScienceDaily, TechXplore 2024

Young Researcher at 10th Heidelberg Laureate Forum | 11% acceptance rate | Link 2023
— *A prestigious forum inviting 200 exceptional young researchers in mathematics and CS worldwide.*

Best Ph.D. Forum Presentation Award at ACM SenSys 2022

Computing Research Association Education **(CRA-E) Graduate Fellow** | Two-year term | Link 2022
— *National committee offers a graduate student role advocating undergraduate research.*

Three Minute Thesis (3MT) finalist at UMASS | Talk Link 2022

Publications

My work has been published at the top venues in virtual, augmented, and mixed reality (VR/AR/MR), robotics, systems, and machine learning, including IEEE VR, TVCG, IROS, NeurIPS, ACM FAccT, and BuildSys/e-Energy.

[1] **Yasra Chandio**, Momin Ahmed Khan, Khotso Selialia, Luis Antonio Garcia, Joseph DeGol, and Fatima M. Anwar: *A Neurosymbolic Approach To Adaptive Feature Extraction In SLAM*, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024.

[2] **Yasra Chandio**, Victoria Interrante, and Fatima M. Anwar: *Balancing Presence And Safety Using Reaction Time In Mixed Reality*, Workshop at IEEE International Symposium on Mixed and Augmented Reality (SafeAR@ISMAR), 2024.

[3] **Yasra Chandio**, Victoria Interrante, and Fatima M. Anwar: *Human Factors At Play: Understanding The Impact Of Conditioning On Presence And Reaction Time In Mixed Reality*, IEEE Conference on Virtual Reality and 3D User Interfaces (VR), 2024. selected to appear in IEEE TVCG **[Top 12% paper]**

[4] **Yasra Chandio**, Noman Bashir, Tian Guo, Elsa Olivetti, and Fatima M. Anwar: *Scoping The Decarbonization Of Collaborative Mixed Reality*, IEEE International Symposium on Emerging Metaverse (ISEMV), 2024.

[5] **Yasra Chandio**, Noman Bashir, and Fatima M. Anwar: *Stealthy and practical multi-modal attacks on mixed reality tracking*, IEEE International Conference on Artificial Intelligence & extended and Virtual Reality (AIxVR), 2024. **[Best Paper Finalist]**

[6] Khotso Selialia, **Yasra Chandio**, and Fatima Anwar: *Mitigating feature heterogeneity biases in Federated Learning*, ACM Conference on Fairness, Accountability, and Transparency (FAccT), 2024.

[7] Momin Ahmed Khan, **Yasra Chandio**, and Fatima Anwar: *HYDRA-FL: Hybrid Knowledge Distillation for Robust and Accurate Federated Learning*, Annual Conference on Neural Information Processing Systems (NeurIPS), 2024.

[8] **Yasra Chandio**, Noman Bashir, Victoria Interrante, and Fatima M. Anwar: *Investigating the Correlation Between Presence and Reaction Time in Mixed Reality*

[9] Noman Bashir, **Yasra Chandio**, David Irwin, Fatima Anwar, Jeremy Gummeson, and Prashant Shenoy: *Jointly managing electrical and thermal energy in solar- and battery-powered computer systems*, ACM International Conference on Future Energy Systems (e-Energy), 2023.

[10] **Yasra Chandio**, Noman Bashir, and Fatima M. Anwar: *Dataset: HoloSet - A dataset for visual inertial pose tracking in extended reality*, 5th International SenSys/BuildSys Workshop on Data (DATA), 2022.

[11] **Yasra Chandio**: *Towards secure and immersive mixed reality: PhD Forum Abstract*, 20th ACM Conference on Embedded Networked Sensor Systems (SenSys), 2022. **[Best Presentation Award]**

[12] Khotso Selialia, **Yasra Chandio**, and Fatima M. Anwar: *Federated Learning Biases in Heterogeneous Edge-Devices - A Case-Study*, 4th International SenSys/BuildSys Workshop on Challenges in Artificial Intelligence and Machine Learning for Internet of Things (AIChallengeIoT), 2022.

[13] **Yasra Chandio** and Fatima M. Anwar: *Poster: Spatiotemporal Security in Mixed Reality systems*, 18th ACM Conference on Embedded Networked Sensor Systems (SenSys), 2020.

[14] **Yasra Chandio**, Aditya Mishra, and Anand Seetharam: *GridPeaks: Employing Distributed Energy Storage For Grid peak Reduction*, IEEE International Green and Sustainable Computing conference (IGSC), 2019.

[15] **Yasra Chandio**, Jo A gila Bitsch, Affan A. Syed, and Muhammad Hamad Alizai: *Networking Wireless Energy in Embedded Networks*, ACM Transactions on Sensor Networks (TOSN), 2018.

[16] Samar Abbas, Abu Bakar, **Yasra Chandio**, Khadija Hafeez, Ayesha Ali, Tariq M Jadoon, and Muhammad Hamad Alizai: *Inverted HVAC: Greenifying Older Buildings, One Room at a Time*, ACM Transactions on Sensor Networks (TOSN), 2018.

[17] Khadija Hafeez, **Yasra Chandio**, Abu Bakar, Ayesha Ali, Affan A. Syed, Tariq M. Jadoon, and Muhammad Hamad Alizai: *Inverting HVAC for Energy Efficient Thermal Comfort in Populous Emerging Countries*, ACM International Conference on Systems for Energy-Efficient Built Environments (BuildSys), 2017. **[Audience Choice Award]**

[18] M. Hamad Alizai, Qasim Raza, **Yasra Chandio**, Affan A. Syed, and Tariq M. Jadoon: *Simulating Intermittently Powered Embedded Networks*, 13th International Conference on Embedded Wireless Systems and Networks (EWSN), 2016.

UNDER-REVIEW/IN-PREPARATION

[1] **Yasra Chandio**, Khotso Selialia, Luis Antonio Garcia, Joseph DeGol, and Fatima M. Anwar: *Lost in Tracking Translation: A Comprehensive Analysis of Visual SLAM in Human-Centered XR and IoT Ecosystems*. **[Preprint]**

[2] **Yasra Chandio**, Victoria Interrante, and Fatima M. Anwar: *Reaction Time as a Proxy for Presence in Mixed Reality with Distraction*. **[Preprint]**

[3] **Yasra Chandio**, Victoria Interrante, and Fatima M. Anwar: *Tap into Reality: Understanding the Impact of Interactions on Presence and Reaction Time in Mixed Reality*. **[Preprint]**

[4] **Yasra Chandio**, Victoria Interrante, and Fatima M. Anwar: *Towards Objective And Implicit Presence Measures In Mixed Reality*. **[Short paper]**

[5] **Yasra Chandio**, Luis Antonio Garcia, Joseph DeGol, and Fatima M. Anwar: *ContinuumTrack: Adaptive Real-Time Profiling and Correction in Dynamic SLAM Systems*. **[in preparation]**

[6] Khotso Selialia, **Yasra Chandio**, Jimi Oke, and Fatima Anwar: *LipFed: Mitigating Subgroup Bias in Federated Learning with Lipschitz Constraints*. **[Preprint]**

[7] Khotso Selialia, **Yasra Chandio**, Jimi Oke, and Fatima Anwar: *Tackling Group Fairness Inconsistencies in Federated Learning*. **[in preparation]**

[8] Diana Romero*, **Yasra Chandio***, Fatima Anwar, and Salma Elmalaki: *GroupBeaMR: Analyzing Collaborative Group Behavior in Mixed Reality Through Passive Sensing and Sociometry*. [* these authors contributed equally] **[Preprint]**

[9] John Murray, **Yasra Chandio**, Fatima Anwar, and Michael Zink: *Exploring Relationship Between Quality Of Experience And Quality Of Service In Collaborative Mixed Reality*. **[in preparation]**

[10] Momin Ahmad Khan, Virat Shejwalkar, **Yasra Chandio**, Amir Houmansadr, and Fatima M. Anwar: *A Critical Analysis Of The Pitfalls In Federated Learning Defense Evaluations*. **[under review]**

[11] Abdullah Al Ishtiaq, Raja Hasnain Anwar, **Yasra Chandio**, Fatima M. Anwar, Syed Rafiul Hussain, and Muhammad Taqi Raza: *Cloud Nine Connectivity: Evading in-Flight Paywalls For Free Internet*. **[under review]**

Open Source Research Artifact

HoloSet - A Dataset for Visual-Inertial Pose Estimation in Extended Reality | Link

Awards and Honors

Nokia Bell Labs Gift for Society for Women Researchers | $2000	2024
UMASS Campus Climate Improvement Grant | $2500	2024
Scholar at CRA-WP Grad Cohort Workshop for IDEALS	2024
IEEE Robotics and Automation Society (RAS) travel grant to attend IROS	2024
Abbe Grant to attend HLF | Highly selective grant, given to 30/200 HLF attendees	2023
CPS-IoT Week SIGBED student travel award | declined	2023
Scholar at CRA-WP Grad Cohort	2022
Invited to attend Google PhD Fellowship Summit	2021
Honorable mention in UMASS ECE 3MT	2021
Selected for Computer Research Mentorship (CSRMP) Program at Google	2021
Grace Hopper Scholar to attend Grace Hopper Celebration	2019
National Science Foundation travel grant to attend IEEE IGSC	2019
Summer Provost Fellowship at Binghamton University	2019
Best Paper Audience Choice Award at ACM BuildSys	2017
ACM SenSys Travel grant to attend ACM BuildSys/SenSys	2017

Media Coverage

Can AR and VR finally disrupt the exhausting culture of video meetings?	BBC
ECE's Yasra Chandio Qualifies as a “Rising Star” in Both Actual and Virtual Reality	CoE
College of Engineering Selects LaChance and Chandio for Its Diversity, Equity, and Inclusion Award	CoE
CRA Education Committee Selects New Graduate Fellow	CRA
New Study: Immersive Engagement In Mixed Reality Can Be Measured With Reaction Time	UMASS
Immersive Engagement in Mixed Reality Can Be Measured with Reaction Time	ScienceDaily
Study: Immersive Engagement in Mixed Reality Can Be Measured with Reaction Time	TechXplore

Teaching and Mentorship

Teaching Assistant

ECE 524/624: Advanced Embedded Systems (Graduate) – helped design course projects — Fall 21-23
ECE 322: Systems Programming — Fall 24
ECE 231: Introduction to Embedded System – helped design new labs — Spring 21-24
ECE 571 (SUNY): Programming Languages (Graduate) — Fall 19
CS 677 (LUMS): Internet of Things — Fall 16

Instructor

Taught Mixed Reality, Programming and helped design Capstone Projects (UMASS Turing Program) — 2022-2025
Embedded System Programming (UMASS 3-week K-12 Engineering Outreach Summer Program) — 2021

Guest Lecturer

Mixed Reality Beyond Pokemon Go (Pioneer Valley Chinese Immersion Charter School) — 2024
Neurosymbolic Human Pose Tracking (CS+Robotics Club at Nashua High School, New Hampshire) — 2024
Deep Learning based Tracking for Mobile Systems (UMASS ECE 524/624) — 2023
Mixed Reality Security and Privacy (UMASS ECE 524/624) — 2022
Introduction to Embedded System with *BeagleBone Black* (UMASS ECE 231) — 2020

Mentorship

John Murray (UMASS PhD) — 2024-
Diana Romero (UC Irvine PhD) — 2023-
Standly Laguerre | NSF REU (Macalester College) — 2023
— *Project: Unpacking Visual SLAM: A Critical Analysis of Tracking Methodologies and Their Pitfalls*
John Dale, Dani Kasti(UMASS BS) — 2022
— *Project: Objectively Measuring Multi-User Collaboration in Mixed Reality*
Kenneth Lin, Yav Rohatgi (UMASS BS) — 2023
— *Project: Remote Interactions in Mixed Reality*
Colin Leydon, Eyado Ocho (UMASS BS) — 2023
— *Project: Deep Learning Based Tracking in Mobile Tracking Systems*
Yifan Meng, Linghao Meng, Shaoyu Dai, Yanan Zhang (UMASS MS) — 2022
— *Project: Pose Estimation for Micro Movements in Mixed Reality*
John Murray, Zachary Wannie, Alexander Dickopf, Zhehang Zhang (UMASS MS) — 2021
— *Project: Deep Learning Based Sensor Fusion for Mixed Reality systems*
Subhankar Chowdhury, Abhinaba Das (UMASS MS) — 2021
— *Project: Deep Learning Based Sensor Fusion for IoT systems*
Xiaohao Xia, Yinxuan Wu, Xintao Ding, Xin Zhao (UMASS MS) — 2021
— *Project: Remote collaboration in Virtual Reality*
Nikhil Sarecha, Rehmat Singh Kang (UMASS MS) — 2020
— *Project: Remote collaboration in Mixed Reality with IoT devices*

Service

Leadership

Chair, College of Engineering Dean's Graduate Advisory Group 2024–2025

Organizer, UMASS ECE Women Dinner Series | UMASS Campus Climate Grant ($2500) | [Link] 2024–2025

— *Community building through a series of dinners with workshops and panels.*

Founding Chair, ECE Grad Gender Diversity group 2023–2025

Inaugural Graduate Student Inductee in ECE DEI Committee 2023–2025

Organizer, ECE Grad Women Retreats | Raised $2500 2023–2025

Organizer and Panelist, ECE Graduate Orientation 2023–2025

Editor, CRA-Undergraduate Research Highlights | [Link] 2022–2024

— *The series features outstanding undergraduate research across North America, offering insights into research experiences, mentorship, career advice, and promoting graduate education.*

Service to the Profession

Technical Program Committee Chair at Sensors S&P@SenSys 2025

Technical Program Committee member at DATA@Buildsys 2024

Technical Program Committee member at ACM S3@Mobicom 2024

Reviewer at IEEE Conference on Virtual Reality And 3D User Interfaces (IEEE VR) 2025

Reviewer at ACM Conference on Human Factors in Computing Systems (ACM CHI) 2025

Reviewer at IEEE International Conference on Robotics & Automation (IEEE ICRA) 2024

Reviewer at International Conference on Learning Representations (ICLR) 2024

Reviewer at Journal of Parallel and Distributed Computing (JPDC) 2024

Shadow Program Committee member at ACM SenSys 2022

Shadow Program Committee member at ACM EuroSys 2022

Review Committee member at Extended Reality track at Grace Hopper Celebration (GHC) 2022

Student Volunteer at ACM SenSys 2022

Review Committee member at VR/AR/MR track at GHC 2021

Panelist

Research Career and Graduate School (UMASS ECE) 2024

— *A panel discussion on academic careers and life at graduate school.*

Surviving PhD (UMASS Graduate Women In STEM) 2024

— *A panel discussion connecting UMASS graduate and undergraduate students to discuss careers in STEM.*

Organizer & Moderator, Life at UMASS as a CS Undergraduate Student (UMASS Turing Program) 2023

Organizer & Moderator, UMASS Women in Engineering Day 2023

— *A panel discussion connecting industry with high school students to discuss careers in engineering.*

Pathways To Graduate Research (UMASS ECE) 2023

— *A panel discussion focused on graduate school experiences of female and non-binary students in ECE*

Service to the Community

Volunteer, Amherst Survival Center 2020

Volunteer, Binghamton University Commencement 2019

Employment

Graduate Research Assistant, UMASS Amherst | *Advisor: Fatima Anwar* `2020-on`

Graduate Research Assistant, Binghamton University | *Advisor: Gaunhua Yan* `2018-2019`

Research Associate, LUMS, Pakistan | *Advisor: Muhammad Hamad Alizai* `2015-2017`

Research Engineer, NUCES, Pakistan | Advisor: *Affan A. Syed* `2014-2015`

References

1. **Fatima Anwar**, Assistant Professor at University of Massachusetts Amherst fanwar@umass.edu
2. **Victoria Interrante**, Professor at University of Minnesota, Twin Cities interran@umn.edu
3. **Michael Zink**, Professor at University of Massachusetts Amherst mzink@umass.edu
4. **Luis Antonio Garcia**, Assistant Professor at University of Utah la.garcia@utah.edu
5. **Joseph DeGol**, CTO at Steg AI jomnipotent17@gmail.com